London, July 10, 2026

Re: RL Fellows Program

To whom it may concern at *Anthropic*,

I am writing to apply for the **RL Fellows Program** at **Anthropic**. My research sits at the intersection of deep RL and large language models, with a focus on **credit assignment** and **scalable RL environments**. I combine an academic track record in RL/LLMs with five years of industry experience shipping ML systems, and I am motivated by the challenge of **scaling RL** to produce safe, capable AI.

Academic and research experience. I am an **Assistant Professor in Machine Learning at UCL** and **Associate Faculty** at the *BOLD Lab* — a £60M UKRI-funded national AI research lab (Oxford, UCL, Imperial) led by Jakob Foerster, focused on open-ended learning and embodied AI. I lead three modules, teach in several others, and have been nominated for the **UCL Inspiring Teaching Delivery Award** two years running.

My **PhD** at UCL, with Laura Toni and Tim Rocktäschel, focused on the Credit Assignment Problem at the intersection of deep RL and LLMs. I authored a **TMLR Survey Certification**-awarded survey on temporal credit assignment, showed that LLMs align with human evaluations of action relevance in RL, built **NAVIX** — a widely adopted JAX reimplementation of MiniGrid at thousands of parallel environments per GPU — and co-authored **BALROG** (ICLR 2025), a benchmark for LLM/VLM agents on long-horizon tasks.

Research agenda. My current work develops **latent option models** that scale options — temporally extended actions — by learning a reusable vocabulary of latent actions from large-scale, **action-free** demonstrations, which can then be fine-tuned with RLHF/RLAIF. This connects directly to the Fellowship's workstreams: multi-turn reasoning and long-horizon tool use in Claude require RL signals that propagate meaningfully across hundreds of steps — the harder the credit assignment problem, the more principled the training scheme needs to be.

Industry experience. Before academia, I spent **five years** translating ML research into commercial products. As **ML Lead** at Burohappold Engineering (2017–2020, London), I delivered deep-learning tools now used by the **Premier League**, creating new revenue streams. At Gridshell.it, open-source genetic algorithms for structural engineering received the **Peter Lord Award**. These roles gave me a **systems mindset** — reproducible codebases and infrastructure in JAX/PyTorch for long-running experiments — and the ability to align scientists, engineers, and stakeholders around ambitious deliverables.

Why the RL Fellows Program at Anthropic? Anthropic is doing the most **ambitious RL-for-alignment** work in the field, at a scale no academic group can match. The Fellowship offers direct mentorship from researchers defining the frontier of post-training, the compute to carry out the experiments for **latent option models** that would compress a year of progress into four months, and a **clear path** to contributing to Anthropic's mission long-term and full-time. My agenda — principled environment design and temporally extended action learning — aligns closely with the goal of improving Claude's capabilities through RL, and I look forward to discussing how I can contribute.

Sincerely,
Eduardo Pignatelli

